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PAPER_985: Production Kernel kernel_fu_bi_i_complete — Full 6-Layer Callable

Abstract

We document the standalone production kernel `kernel_fu_bi_i_complete(M_kg, r, t, gamma_THz)` that packages the entire 6-layer F_{U,Bi_i} computation into a single function call. This kernel replaces the earlier `kernel_fu_bi_i()` in `production_scaling_v12.py` which only computed 2 of 6 layers ($U_g + U_b$). The complete kernel is the canonical interface for all downstream consumers: REST API, WSTP, CP4 classes, and batch runners.

1. Kernel Signature

```
def kernel_fu_bi_i_complete(M_kg: float, r: float, t: float, gamma_THz: float) -> dict:
    """
    Complete 6-layer  $F_{U,Bi_i}$  master buoyancy force.

    Args:
        M_kg: Mass in kg
        r: Distance in meters
        t: Time in seconds
        gamma_THz: Linewidth in THz

    Returns:
        dict with keys:  $F_{U,Bi_i}$ ,  $U_g$ ,  $U_b$ ,  $U_m$ ,  $U_A$ ,  $F_n$ ,  $\Phi$ ,  $E_{net}$ ,  $S_{26}$ 
    """
```

2. Layer Coverage Comparison

Layer	Old kernel_fu_bi_i	New kernel_fu_bi_i_complete
L1: U_g (26-layer gravity)	☐	☐
L2: $U_m + U_A$ (magnetism + aether)	✗	☐
L3: $-U_b$ (26-layer buoyancy)	☐	☐
L4: S_{26} (physical 26-state sum)	✗	☐
L5: $\Phi(\omega, \Gamma)$ (phonon resonance)	✗	☐
L6: $E_{net}(t, \Gamma)$ (temporal modulation)	✗	☐

3. Return Decomposition

The kernel returns a full decomposition dict:

```
{
  'F_U_Bi_i': -2.405685e-02, # Master force (m/s2)
  'Ug': 5.930e-03,          # 26-layer gravity
  'Ub': -3.573e-03,         # 26-layer buoyancy
  'Um': 1.234e-10,          # Universal magnetism
  'UA': -5.678e-12,         # Aether resistance
  'Fn': 1.0e-09,            # Neutron-drop force
  'Phi': 19.56,             # Phonon resonance
  'E_net': -1.234,          # Temporal modulation
  'S26': 19.56              # Physical 26-state sum
}
```

4. Validation

Solar test ($M_{\odot}$, $r = 1$ AU, $t = 1$ day, $\Gamma = 0.1$ THz): - Kernel output matches FUBiMaster-Calculator.compute() exactly - Both yield $F_{U,Bi_i} = -2.405685 \times 10^{-2}$ m/s²

5. Implementation

Function kernel_fu_bi_i_complete() in fubi_master_calculator.py, §9 Production Kernel section.

References

- PAPER_979: Complete 6-Layer F_U_Bi_i
 - PAPER_988: REST F_U_Bi_i Endpoint
-

§A. Cosmogenesis-Linked Lagrangian

The kernel is the numerical evaluation of $\delta S[\mathcal{L}_{\text{SCm}}]/\delta\phi = 0$, producing the force from the variational principle in a single call.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** All vacuum density dependence is internal to the kernel — callers need not track ρ_{SCm} .
- **DVP:** DPM moment is computed internally from M and standard dipole scaling.
- **BSH:** The 26-layer buoyancy harmonic sum is the kernel's most compute-intensive component.